INTRODUCTION TO PROJECT

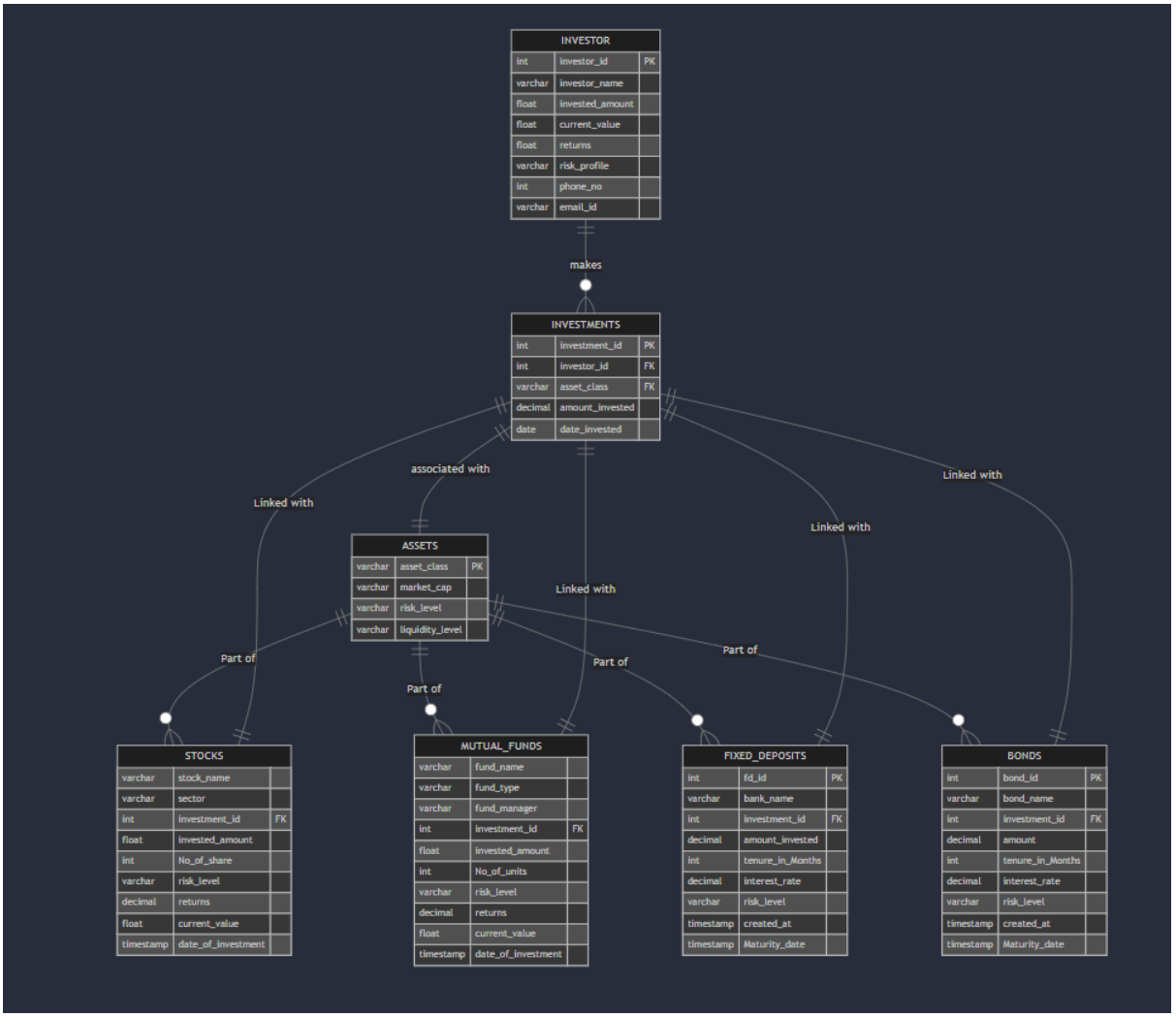
In today's dynamic financial landscape, managing an investment portfolio efficiently is crucial for achieving long-term financial goals. This project, titled *Investment Portfolio Management System*, aims to simplify the complex process of tracking, managing, and analyzing diverse asset classes. The portfolio includes multiple investment types such as stocks, mutual funds, bonds, and fixed deposits, each with unique characteristics and risk profiles.

The system is designed to store comprehensive information about each investment category, enabling users to gain insights into their asset allocations and make informed decisions. This structured approach allows users to keep track of individual investment details like invested amount, risk level, returns, and maturity dates, while also ensuring that the total invested capital aligns across various asset classes.

This project uses SQL databases to maintain data consistency, relational integrity, and easy retrieval of investment data. The design incorporates relationships between entities such as investors, investments, and assets to provide a robust, interconnected view of the portfolio. By integrating asset class-specific tables, users can manage their investments with precision, optimize their returns, and gain a clearer understanding of their financial positions.

This system is especially beneficial for individuals and financial analysts looking to monitor investment performance over time, make data-driven decisions, and ensure their portfolios remain balanced and aligned with their financial goals.

ER-DIAGRAM



Entities:

- Investors
- Investments
- Assets
- Stocks
- Mutual_Funds
- Bonds
- Fixed_Deposits

Significance of Each Table:

1. **ASSETS:** The `ASSETS` table serves as a catalog of different types of financial assets, categorized by asset type, class, risk, and liquidity level. This table acts as a reference point for linking specific financial instruments like stocks, bonds, mutual funds, and fixed deposits. It aids in classifying assets and understanding their associated risk and liquidity characteristics.
2. **INVESTORS:** The `INVESTORS` table stores essential information about each investor, including their personal details and financial status in terms of invested amounts, current portfolio value, and returns. This table forms the core of the database as each investor can be linked to multiple investments, which are tracked across various asset classes.
3. **INVESTMENTS:** The `INVESTMENTS` table logs every investment made by investors. It links the investor to the specific asset in which they have invested, along with the investment details such as the amount invested and the date of investment. This table plays a pivotal role in consolidating and managing all investment activities across asset classes.
4. **STOCKS:** The `STOCKS` table captures data related to individual stocks, including their names, associated stock exchange, sector, market capitalization, risk level, return rate, and current share price. This table is essential for managing equity investments and assessing stock market exposure.
5. **MUTUAL_FUNDS:** The `MUTUAL_FUNDS` table stores details of mutual funds such as their names, types, fund manager, net asset value (NAV), risk levels, and return rates. It is crucial for understanding the mutual fund segment of the portfolio and for selecting appropriate funds based on risk tolerance.
6. **BONDS:** The `BONDS` table contains detailed information about different bonds, such as bond names, types, interest rates, maturity periods, face values, and risk levels. This table is significant for managing fixed-income investments and understanding the duration and yield of bond holdings.

7. **FIXED_DEPOSITS:** The `FIXED_DEPOSITS` table records information related to fixed deposits, including the bank name, interest rates, tenure, and amount. This table is crucial for managing low-risk, interest-bearing investments in fixed deposits.

Table Schema Definitions

1. INVESTOR Table

- **investor_id**: `INT` — Unique identifier for each investor (Primary Key).
- **investor_name**: `VARCHAR(255)` — Name of the investor.
- **invested_amount**: `FLOAT` — Total amount invested by the investor.
- **current_value**: `FLOAT` — Current value of the investor's total investments.
- **returns**: `FLOAT` — Calculated returns on the investor's investments.
- **risk_profile**: `VARCHAR(100)` — Investor's risk profile (e.g., Low, Medium, High).
- **phone_no**: `BIGINT` — Investor's contact number.
- **email_id**: `VARCHAR(255)` — Investor's email address.

2. ASSETS Table

- **asset_class**: `VARCHAR(50)` — Name/type of the asset class (Primary Key).
- **market_cap**: `FLOAT` — Market capitalization of the asset class.
- **risk_level**: `VARCHAR(50)` — Risk level associated with the asset class.
- **liquidity_level**: `VARCHAR(50)` — Liquidity level of the asset class.

3. INVESTMENTS Table

- **investment_id**: `INT` — Unique identifier for each investment (Primary Key).
- **investor_id**: `INT` — ID of the investor making the investment (Foreign Key references `INVESTOR`).
- **asset_class**: `VARCHAR(50)` — Class/type of asset (Foreign Key references `ASSETS`).
- **amount_invested**: `DECIMAL(15, 2)` — Amount invested in the specific investment.
- **date_invested**: `DATE` — Date when the investment was made.

4. STOCKS Table

- **stock_name**: `VARCHAR(255)` — Name of the stock.
- **sector**: `VARCHAR(100)` — Sector the stock belongs to (e.g., Technology, Healthcare).
- **investment_id**: `INT` — Reference to the investment ID (Foreign Key references `INVESTMENTS`).
- **invested_amount**: `FLOAT` — Amount invested in this specific stock.
- **no_of_share**: `INT` — Number of shares bought.
- **risk_level**: `VARCHAR(50)` — Risk level associated with the stock.
- **returns**: `DECIMAL(15, 2)` — Returns on this stock investment.
- **current_value**: `FLOAT` — Current value of the stock investment.
- **date_of_investment**: `TIMESTAMP` — Date and time of the stock investment.

5. MUTUAL_FUNDS Table

- **fund_name**: `VARCHAR(255)` — Name of the mutual fund.
- **fund_type**: `VARCHAR(100)` — Type of mutual fund (e.g., Equity, Debt).
- **fund_manager**: `VARCHAR(100)` — Name of the fund manager.
- **investment_id**: `INT` — Reference to the investment ID (Foreign Key references `INVESTMENTS`).
- **invested_amount**: `FLOAT` — Amount invested in this mutual fund.
- **no_of_units**: `FLOAT` — Number of units bought in the mutual fund.
- **risk_level**: `VARCHAR(50)` — Risk level of the mutual fund.
- **returns**: `DECIMAL(15, 2)` — Returns on this mutual fund investment.
- **current_value**: `FLOAT` — Current value of the mutual fund investment.
- **date_of_investment**: `TIMESTAMP` — Date and time of the mutual fund investment.

6. FIXED_DEPOSITS Table

- **fd_id**: `INT` — Unique identifier for each fixed deposit (Primary Key).
- **bank_name**: `VARCHAR(255)` — Name of the bank where the FD is held.
- **investment_id**: `INT` — Reference to the investment ID (Foreign Key references `INVESTMENTS`).
- **amount_invested**: `DECIMAL(15, 2)` — Amount invested in the fixed deposit.
- **tenure_in_months**: `INT` — Tenure of the FD in months.
- **interest_rate**: `DECIMAL(5, 2)` — Interest rate offered on the FD.
- **risk_level**: `VARCHAR(50)` — Risk level of the FD.
- **created_at**: `TIMESTAMP` — Date and time of FD creation (default to the current timestamp).
- **maturity_date**: `TIMESTAMP` — Date when the FD matures.

7. BONDS Table

- **bond_id**: `INT` — Unique identifier for each bond (Primary Key).
- **bond_name**: `VARCHAR(255)` — Name of the bond.
- **investment_id**: `INT` — Reference to the investment ID (Foreign Key references `INVESTMENTS`).
- **invested_amount**: `DECIMAL(15, 2)` — Amount invested in this bond.
- **tenure_in_months**: `INT` — Tenure of the bond in months.
- **interest_rate**: `DECIMAL(5, 2)` — Interest rate of the bond.
- **risk_level**: `VARCHAR(50)` — Risk level of the bond.
- **created_at**: `TIMESTAMP` — Date and time the bond was created (default to the current timestamp).
- **maturity_date**: `TIMESTAMP` — Maturity date of the bond.

COMMANDS:

1. Create the Database:

CREATE DATABASE Investment Portfolio Management;
USE Investment Portfolio Management;

CREATING TABLES:

```
-- Create INVESTOR Table
CREATE TABLE INVESTOR (
   investor_id INT PRIMARY KEY,
   investor_name VARCHAR(255),
   invested_amount FLOAT,
   current_value FLOAT,
   returns FLOAT,
   risk_profile VARCHAR(100),
   phone_no BIGINT,
   email_id VARCHAR(255)
);

-- Create ASSETS Table
CREATE TABLE ASSETS (
   asset_class VARCHAR(50) PRIMARY KEY,
   market_cap float,
   risk_level VARCHAR(50),
   liquidity_level VARCHAR(50)
);

-- Create INVESTMENTS Table
CREATE TABLE INVESTMENTS (
   investment_id INT PRIMARY KEY,
   investor_id INT,
   asset_class VARCHAR(50),
   amount_invested DECIMAL(15, 2),
   date_invested DATE,
   FOREIGN KEY (investor_id) REFERENCES INVESTOR(investor_id),
   FOREIGN KEY (asset_class) REFERENCES ASSETS(asset_class)
);

-- Create STOCKS Table
CREATE TABLE STOCKS (
   stock_name VARCHAR(255),
   sector VARCHAR(100),
   investment_id INT,
   invested_amount FLOAT,
   no_of_share INT,
   risk_level VARCHAR(50),
   returns DECIMAL(15, 2),
   current_value FLOAT,
```

```
    date_of_investment TIMESTAMP,
    FOREIGN KEY (investment_id) REFERENCES INVESTMENTS(investment_id)
);

-- Create MUTUAL_FUNDS Table
CREATE TABLE MUTUAL_FUNDS (
    fund_name VARCHAR(255),
    fund_type VARCHAR(100),
    fund_manager VARCHAR(100),
    investment_id INT,
    invested_amount FLOAT,
    no_of_units Float,
    risk_level VARCHAR(50),
    returns DECIMAL(15, 2),
    current_value FLOAT,
    date_of_investment TIMESTAMP,
    FOREIGN KEY (investment_id) REFERENCES INVESTMENTS(investment_id)
);

-- Create FIXED_DEPOSITS Table
CREATE TABLE FIXED_DEPOSITS (
    fd_id INT PRIMARY KEY,
    bank_name VARCHAR(255),
    investment_id INT,
    amount_invested DECIMAL(15, 2),
    tenure_in_months INT,
    interest_rate DECIMAL(5, 2),
    risk_level VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    maturity_date TIMESTAMP,
    FOREIGN KEY (investment_id) REFERENCES INVESTMENTS(investment_id)
);

-- Create BONDS Table
CREATE TABLE BONDS (
    bond_id INT PRIMARY KEY,
    bond_name VARCHAR(255),
    investment_id INT,
    invested_amount DECIMAL(15, 2),
    tenure_in_months INT,
    interest_rate DECIMAL(5, 2),
    risk_level VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    maturity_date TIMESTAMP,
    FOREIGN KEY (investment_id) REFERENCES INVESTMENTS(investment_id)
);
```

INSERTING VALUES IN ALL THE TABLES:

Insert into Investor

```
INSERT INTO INVESTOR (investor_id, investor_name, invested_amount, current_value,
returns, risk_profile, phone_no, email_id)
VALUES
(1, 'Aarav Sharma', 1000000, 6000000, 500.0, 'High', 9876543210,
'aarav.sharma@example.com'),
(2, 'Vivaan Patel', 500000, 475000, -5.0, 'Medium', 9876543211,
'vivaan.patel@example.com'),
(3, 'Aditya Verma', 750000, 1300000, 73.33, 'Low', 9876543212,
'aditya.verma@example.com'),
(4, 'Sai Kumar', 300000, 900000, 200.0, 'High', 9876543213, 'sai.kumar@example.com'),
(5, 'Reyansh Gupta', 600000, 900000, 50.0, 'Medium', 9876543214,
'reyansh.gupta@example.com'),
(6, 'Krishna Iyer', 200000, 210000, 5.0, 'Low', 9876543215,
'krishna.iyer@example.com'),
(7, 'Aarohi Singh', 450000, 540000, 20.0, 'Medium', 9876543216,
'aarohi.singh@example.com'),
(8, 'Kabir Joshi', 900000, 600000, -33.33, 'High', 9876543217,
'kabir.joshi@example.com'),
(9, 'Diya Reddy', 250000, 340000, 36.0, 'Low', 9876543218, 'diya.reddy@example.com'),
(10, 'Karan Mehta', 400000, 360000, -10.0, 'Medium', 9876543219,
'karan.mehta@example.com'),
(11, 'Simran Kapoor', 700000, 900000, 28.57, 'High', 9876543220,
'simran.kapoor@example.com'),
(12, 'Rohit Sethi', 350000, 380000, 8.57, 'Low', 9876543221,
'rohit.sethi@example.com'),
(13, 'Neha Bansal', 550000, 770000, 40.0, 'Medium', 9876543222,
'neha.bansal@example.com'),
(14, 'Ananya Nair', 800000, 1600000, 100.0, 'High', 9876543223,
'ananya.nair@example.com'),
(15, 'Nishant Choudhury', 300000, 315000, 5.0, 'Low', 9876543224,
'nishant.choudhury@example.com'),
(16, 'Riya Saxena', 650000, 450000, -30.77, 'Medium', 9876543225,
'riya.saxena@example.com'),
(17, 'Tarun Rao', 200000, 240000, 20.0, 'High', 9876543226, 'tarun.rao@example.com'),
(18, 'Siddharth Rani', 400000, 410000, 2.5, 'Low', 9876543227,
'siddharth.rani@example.com'),
(19, 'Ravi Yadav', 900000, 2000000, 122.22, 'Medium', 9876543228,
'ravi.yadav@example.com'),
```

(20, 'Pooja Jain', 500000, 1100000, 120.0, 'High', 9876543229,
'pooja.jain@example.com'),
(21, 'Kavya Bhattacharya', 250000, 300000, 20.0, 'Low', 9876543230,
'kavya.bhattacharya@example.com'),
(22, 'Mohit Ghosh', 700000, 740000, 5.71, 'Medium', 9876543231,
'mohit.ghosh@example.com'),
(23, 'Sanya Malhotra', 600000, 620000, 3.33, 'High', 9876543232,
'sanya.malhotra@example.com'),
(24, 'Anuj Khanna', 350000, 370000, 5.71, 'Low', 9876543233,
'anuj.khanna@example.com'),
(25, 'Tanya Agarwal', 800000, 840000, 5.0, 'Medium', 9876543234,
'tanya.agarwal@example.com'),
(26, 'Vikas Soni', 300000, 315000, 5.0, 'High', 9876543235, 'vikas.soni@example.com'),
(27, 'Priya Sharma', 450000, 470000, 4.44, 'Low', 9876543236,
'priya.sharma@example.com'),
(28, 'Ajay Bhatia', 550000, 580000, 5.45, 'Medium', 9876543237,
'ajay.bhatia@example.com'),
(29, 'Chaitanya Tripathi', 700000, 800000, 14.29, 'High', 9876543238,
'chaitanya.tripathi@example.com'),
(30, 'Kriti Patel', 400000, 440000, 10.0, 'Low', 9876543239, 'kriti.patel@example.com'),
(31, 'Gaurav Joshi', 600000, 700000, 16.67, 'Medium', 9876543240,
'gaurav.joshi@example.com'),
(32, 'Tanvi Sahu', 250000, 350000, 40.0, 'High', 9876543241,
'tanvi.sahu@example.com'),
(33, 'Bharat Dubey', 500000, 1000000, 100.0, 'Low', 9876543242,
'bharat.dubey@example.com'),
(34, 'Sneha Singh', 300000, 310000, 3.33, 'Medium', 9876543243,
'sneha.singh@example.com'),
(35, 'Harman Thakur', 400000, 380000, -5.0, 'High', 9876543244,
'harman.thakur@example.com'),
(36, 'Neeraj Ahuja', 600000, 630000, 5.0, 'Low', 9876543245,
'neeraj.ahuja@example.com'),
(37, 'Rohan Khanna', 900000, 750000, -16.67, 'Medium', 9876543246,
'rohan.khanna@example.com'),
(38, 'Aditi Gupta', 700000, 1000000, 42.86, 'High', 9876543247,
'aditi.gupta@example.com'),
(39, 'Vijay Reddy', 600000, 780000, 30.0, 'Low', 9876543248,
'vijay.reddy@example.com'),
(40, 'Shivani Yadav', 800000, 400000, -50.0, 'Medium', 9876543249,
'shivani.yadav@example.com'),
(41, 'Amit Sethi', 600000, 640000, 6.67, 'High', 9876543250, 'amit.sethi@example.com'),
(42, 'Diksha Jain', 250000, 260000, 4.0, 'Low', 9876543251,
'diksha.jain@example.com'),
(43, 'Suraj Singh', 400000, 410000, 2.5, 'Medium', 9876543252,
'suraj.singh@example.com'),

(44, 'Parul Sharma', 500000, 480000, -4.0, 'High', 9876543253, 'parul.sharma@example.com'),
(45, 'Kabir Yadav', 350000, 350000, 0.0, 'Low', 9876543254, 'kabir.yadav@example.com'),
(46, 'Vinay Choudhury', 900000, 1000000, 11.11, 'Medium', 9876543255, 'vinay.choudhury@example.com'),
(47, 'Neha Gupta', 300000, 290000, -3.33, 'High', 9876543256, 'neha.gupta@example.com'),
(48, 'Ananya Khanna', 450000, 460000, 2.22, 'Low', 9876543257, 'ananya.khanna@example.com'),
(49, 'Rohit Mehta', 800000, 850000, 6.25, 'Medium', 9876543258, 'rohit.mehta@example.com'),
(50, 'Krishna Reddy', 200000, 300000, 50.0, 'Medium', 9876543215, 'krishna.reddy@example.com'),
(51, 'Meera Joshi', 450000, 675000, 50.0, 'Low', 9876543216, 'meera.joshi@example.com'),
(52, 'Arjun Nair', 900000, 1800000, 100.0, 'High', 9876543217, 'arjun.nair@example.com'),
(53, 'Isha Singh', 250000, 262500, 5.0, 'Medium', 9876543218, 'isha.singh@example.com'),
(54, 'Devansh Roy', 400000, 460000, 15.0, 'Low', 9876543219, 'devansh.roy@example.com'),
(55, 'Aditi Desai', 700000, 805000, 15.0, 'Medium', 9876543220, 'aditi.desai@example.com'),
(56, 'Rohan Mishra', 350000, 490000, 40.0, 'High', 9876543221, 'rohan.mishra@example.com'),
(57, 'Naina Agarwal', 550000, 770000, 40.0, 'Medium', 9876543222, 'naina.agarwal@example.com'),
(58, 'Kavya Mehta', 800000, 1040000, 30.0, 'Low', 9876543223, 'kavya.mehta@example.com'),
(59, 'Aarohi Chauhan', 300000, 420000, 40.0, 'High', 9876543224, 'aarohi.chauhan@example.com'),
(60, 'Tanmay Singh', 650000, 845000, 30.0, 'Medium', 9876543225, 'tanmay.singh@example.com'),
(61, 'Pranav Sharma', 200000, 270000, 35.0, 'Low', 9876543226, 'pranav.sharma@example.com'),
(62, 'Saanvi Rao', 400000, 480000, 20.0, 'Medium', 9876543227, 'saanvi.rao@example.com'),
(63, 'Ayaan Malik', 900000, 1350000, 50.0, 'High', 9876543228, 'ayaan.malik@example.com'),
(64, 'Ritika Kapoor', 500000, 575000, 15.0, 'Low', 9876543229, 'ritika.kapoor@example.com');

OUTPUT:

investor_id	investor_name	invested_amount	current_value	returns	risk_profile	phone_no	email_id
1	Aarav Sharma	1000000	6000000	500	High	9876543210	aarav.sharma@example.com
2	Vivaan Patel	500000	475000	-5	Medium	9876543211	vivaan.patel@example.com
3	Aditya Verma	750000	1300000	73.33	Low	9876543212	aditya.verma@example.com
4	Sai Kumar	300000	900000	200	High	9876543213	sai.kumar@example.com
5	Reyansh Gupta	600000	900000	50	Medium	9876543214	reyansh.gupta@example.com
6	Krishna Iyer	200000	210000	5	Low	9876543215	krishna.iyer@example.com
7	Aarohi Singh	450000	540000	20	Medium	9876543216	aarohi.singh@example.com
8	Kabir Joshi	900000	600000	-33.33	High	9876543217	kabir.joshi@example.com
9	Diya Reddy	250000	340000	36	Low	9876543218	diya.reddy@example.com
10	Karan Mehta	400000	360000	-10	Medium	9876543219	karan.mehta@example.com
11	Simran Kapoor	700000	900000	28.57	High	9876543220	simran.kapoor@example.com

INSERT INTO INVESTMENTS

INSERT INTO INVESTMENTS (investment_id, investor_id, asset_class, amount_invested, date_invested)
VALUES
(1, 1, 'Stocks', 1000000.00, '2020-01-15'),
(2, 2, 'Mutual Funds', 500000.00, '2019-03-11'),
(3, 3, 'Fixed Deposits', 750000.00, '2020-07-18'),
(4, 4, 'Stocks', 300000.00, '2021-04-25'),
(5, 5, 'Mutual Funds', 600000.00, '2022-08-15'),
(6, 6, 'Fixed Deposits', 200000.00, '2021-11-05'),
(7, 7, 'Bonds', 450000.00, '2021-06-12'),
(8, 8, 'Stocks', 900000.00, '2023-01-17'),
(9, 9, 'Mutual Funds', 250000.00, '2020-03-14'),
(10, 10, 'Bonds', 400000.00, '2022-09-25'),
(11, 11, 'Stocks', 700000.00, '2022-10-30'),
(12, 12, 'Fixed Deposits', 350000.00, '2020-12-20'),
(13, 13, 'Mutual Funds', 550000.00, '2023-04-10'),
(14, 14, 'Stocks', 800000.00, '2023-05-25'),
(15, 15, 'Bonds', 300000.00, '2019-02-08'),
(16, 16, 'Mutual Funds', 650000.00, '2021-07-04'),
(17, 17, 'Stocks', 200000.00, '2020-02-12'),
(18, 18, 'Fixed Deposits', 400000.00, '2019-11-19'),
(19, 19, 'Mutual Funds', 900000.00, '2022-01-09'),
(20, 20, 'Stocks', 500000.00, '2021-06-06'),
(21, 21, 'Bonds', 250000.00, '2022-09-12'),
(22, 22, 'Fixed Deposits', 700000.00, '2021-03-03'),
(23, 23, 'Stocks', 600000.00, '2023-02-14'),
(24, 24, 'Bonds', 350000.00, '2020-10-05'),
(25, 25, 'Mutual Funds', 800000.00, '2022-12-20'),
(26, 26, 'Stocks', 300000.00, '2023-04-18'),
(27, 27, 'Fixed Deposits', 450000.00, '2022-05-22'),
(28, 28, 'Mutual Funds', 550000.00, '2021-06-16'),
(29, 29, 'Stocks', 700000.00, '2023-07-23'),
(30, 30, 'Fixed Deposits', 400000.00, '2020-01-10'),
(31, 31, 'Mutual Funds', 600000.00, '2022-09-11'),

(32, 32, 'Stocks', 250000.00, '2020-12-31'),
(33, 33, 'Fixed Deposits', 500000.00, '2021-08-27'),
(34, 34, 'Mutual Funds', 300000.00, '2022-07-01'),
(35, 35, 'Stocks', 400000.00, '2021-09-15'),
(36, 36, 'Bonds', 600000.00, '2020-02-25'),
(37, 37, 'Mutual Funds', 900000.00, '2021-11-09'),
(38, 38, 'Stocks', 700000.00, '2023-04-11'),
(39, 39, 'Fixed Deposits', 600000.00, '2019-06-13'),
(40, 40, 'Mutual Funds', 800000.00, '2022-05-14'),
(41, 41, 'Stocks', 600000.00, '2023-03-20'),
(42, 42, 'Fixed Deposits', 250000.00, '2021-12-07'),
(43, 43, 'Bonds', 400000.00, '2021-04-19'),
(44, 44, 'Stocks', 500000.00, '2023-01-22'),
(45, 45, 'Fixed Deposits', 350000.00, '2020-03-27'),
(46, 46, 'Mutual Funds', 900000.00, '2023-07-30'),
(47, 47, 'Stocks', 300000.00, '2020-05-16'),
(48, 48, 'Bonds', 450000.00, '2022-10-09'),
(49, 49, 'Fixed Deposits', 800000.00, '2021-06-03'),
(50, 50, 'Stocks', 100000.00, '2020-01-15'),
(51, 50, 'Mutual Funds', 50000.00, '2021-06-11'),
(52, 50, 'Bonds', 50000.00, '2022-03-15'),
(53, 51, 'Fixed Deposits', 150000.00, '2021-02-10'),
(54, 51, 'Stocks', 200000.00, '2022-08-05'),
(55, 51, 'Mutual Funds', 100000.00, '2023-01-14'),
(56, 52, 'Stocks', 400000.00, '2020-07-01'),
(57, 52, 'Mutual Funds', 300000.00, '2021-09-23'),
(58, 52, 'Bonds', 200000.00, '2022-11-10'),
(59, 53, 'Fixed Deposits', 150000.00, '2020-04-12'),
(60, 53, 'Stocks', 100000.00, '2023-07-15'),
(61, 54, 'Mutual Funds', 200000.00, '2021-05-21'),
(62, 54, 'Stocks', 200000.00, '2022-11-10'),
(63, 55, 'Stocks', 300000.00, '2020-09-15'),
(64, 55, 'Fixed Deposits', 200000.00, '2021-06-10'),
(65, 55, 'Mutual Funds', 200000.00, '2022-08-19'),
(66, 56, 'Bonds', 150000.00, '2021-11-05'),
(67, 56, 'Stocks', 200000.00, '2023-01-20'),
(68, 57, 'Fixed Deposits', 300000.00, '2020-04-18'),
(69, 57, 'Stocks', 250000.00, '2022-06-15'),
(70, 58, 'Stocks', 500000.00, '2021-07-10'),
(71, 58, 'Bonds', 300000.00, '2023-03-25'),
(72, 59, 'Fixed Deposits', 150000.00, '2020-03-11'),
(73, 59, 'Mutual Funds', 150000.00, '2022-09-09'),
(74, 60, 'Stocks', 350000.00, '2020-05-16'),
(75, 60, 'Mutual Funds', 300000.00, '2022-11-17'),
(76, 61, 'Bonds', 100000.00, '2020-01-11'),
(77, 61, 'Fixed Deposits', 100000.00, '2022-05-22'),

(78, 62, 'Stocks', 250000.00, '2020-08-18'),
(79, 62, 'Mutual Funds', 150000.00, '2023-04-10'),
(80, 63, 'Mutual Funds', 500000.00, '2021-09-22'),
(81, 63, 'Stocks', 400000.00, '2023-06-19'),
(82, 64, 'Fixed Deposits', 300000.00, '2020-10-13'),
(83, 64, 'Stocks', 200000.00, '2022-12-14');

OUTPUT:

investment_id	investor_id	asset_class	amount_invested	date_invested
1	1	Stocks	1000000.00	2020-01-15
2	2	Mutual Funds	500000.00	2019-03-11
3	3	Fixed Deposits	750000.00	2020-07-18
4	4	Stocks	300000.00	2021-04-25
5	5	Mutual Funds	600000.00	2022-08-15
6	6	Fixed Deposits	200000.00	2021-11-05
7	7	Bonds	450000.00	2021-06-12
8	8	Stocks	900000.00	2023-01-17
9	9	Mutual Funds	250000.00	2020-03-14
10	10	Bonds	400000.00	2022-09-25
11	11	Stocks	700000.00	2022-10-30
12	12	Fixed Deposits	350000.00	2020-12-20

INSERT INTO ASSETS

INSERT INTO ASSETS (asset_class, market_cap, risk_level, liquidity_level)
VALUES
('Stocks', 200000000000, 'High', 'Medium'),
('Mutual Funds', 15000000000, 'Medium', 'High'),
('Fixed Deposits', 5000000000, 'Low', 'Low'),
('Bonds', 10000000000, 'Low', 'Medium');

OUTPUT:

asset_class	market_cap	risk_level	liquidity_level
Bonds	1200000000	Low	Low
Fixed Deposits	500000000	Low	Low
Mutual Funds	800000000	Medium	Medium
Stocks	1500000000	High	High

INSERT INTO STOCKS

INSERT INTO STOCKS(stock_name, sector, investment_id, invested_amount, no_of_share, risk_level, returns, current_value, date_of_investment)
VALUES

-- Each stock record below is linked to an investment_id from the investments table, focusing on Indian companies and stocks only
('Reliance Industries', 'Energy', 1, 1000000.00, 100, 'Medium', 150000.00, 1150000.00, '2020-01-15'),
('Tata Consultancy Services', 'IT', 4, 300000.00, 20, 'Low', 60000.00, 360000.00, '2021-04-25'),
('Infosys', 'IT', 8, 900000.00, 50, 'Medium', 180000.00, 1080000.00, '2023-01-17'),
('HDFC Bank', 'Banking', 11, 700000.00, 30, 'Low', 105000.00, 805000.00, '2022-10-30'),
('ICICI Bank', 'Banking', 14, 800000.00, 25, 'Low', 120000.00, 920000.00, '2023-05-25'),
('Asian Paints', 'Consumer Goods', 17, 200000.00, 15, 'Low', 30000.00, 230000.00, '2020-02-12'),
('Bharti Airtel', 'Telecom', 20, 500000.00, 100, 'Medium', 75000.00, 575000.00, '2021-06-06'),
('Larsen & Toubro', 'Infrastructure', 23, 600000.00, 20, 'High', 90000.00, 690000.00, '2023-02-14'),
('Bajaj Finance', 'Finance', 26, 300000.00, 10, 'High', 45000.00, 345000.00, '2023-04-18'),
('State Bank of India', 'Banking', 29, 700000.00, 40, 'Medium', 105000.00, 805000.00, '2023-07-23'),
('Mahindra & Mahindra', 'Automobile', 32, 250000.00, 25, 'Medium', 37500.00, 287500.00, '2020-12-31'),
('UltraTech Cement', 'Construction', 35, 400000.00, 10, 'High', 60000.00, 460000.00, '2021-09-15'),
('HCL Technologies', 'IT', 38, 700000.00, 35, 'Medium', 105000.00, 805000.00, '2023-04-11'),
('Wipro', 'IT', 41, 600000.00, 30, 'Low', 90000.00, 690000.00, '2023-03-20'),
('Axis Bank', 'Banking', 44, 500000.00, 25, 'Medium', 75000.00, 575000.00, '2023-01-22'),
('Maruti Suzuki', 'Automobile', 47, 300000.00, 15, 'High', 45000.00, 345000.00, '2020-05-16'),
('Hindustan Unilever', 'Consumer Goods', 50, 100000.00, 10, 'Low', 15000.00, 115000.00, '2020-01-15'),
('ITC', 'Consumer Goods', 54, 200000.00, 30, 'Low', 30000.00, 230000.00, '2022-08-05'),
('Adani Enterprises', 'Conglomerate', 56, 400000.00, 20, 'High', 60000.00, 460000.00, '2020-07-01'),
('NTPC', 'Energy', 60, 350000.00, 50, 'Medium', 52500.00, 402500.00, '2020-05-16'),
('Coal India', 'Mining', 63, 300000.00, 100, 'Medium', 45000.00, 345000.00, '2020-09-15'),
('Power Grid Corporation', 'Energy', 67, 200000.00, 60, 'Low', 30000.00, 230000.00, '2023-01-20'),
('JSW Steel', 'Steel', 69, 250000.00, 80, 'Medium', 37500.00, 287500.00, '2022-06-15'),
('Reliance Industries', 'Energy', 70, 500000.00, 100, 'Moderate', 10.50, 550000.00, '2021-07-10'),
('Grasim Industries', 'Conglomerate', 74, 350000.00, 40, 'High', 52500.00, 402500.00, '2020-05-16'),

('Tata Steel', 'Steel', 78, 250000.00, 35, 'Medium', 37500.00, 287500.00, '2020-08-18'),
('IndusInd Bank', 'Banking', 81, 400000.00, 20, 'High', 60000.00, 460000.00, '2023-06-19'),
('Sun Pharmaceuticals', 'Pharmaceutical', 83, 200000.00, 25, 'Low', 30000.00, 230000.00, '2022-12-14');

OUTPUT:

stock_name	sector	investment_id	invested_amount	no_of_share	risk_level	returns	current_value	date_of_investment
Reliance Industries	Energy	1	1000000	100	Medium	150000.00	1150000	2020-01-15 00:00:00
Tata Consultancy Services	IT	4	300000	20	Low	60000.00	360000	2021-04-25 00:00:00
Infosys	IT	8	900000	50	Medium	180000.00	1080000	2023-01-17 00:00:00
HDFC Bank	Banking	11	700000	30	Low	105000.00	805000	2022-10-30 00:00:00
ICICI Bank	Banking	14	800000	25	Low	120000.00	920000	2023-05-25 00:00:00
Asian Paints	Consumer Goods	17	200000	15	Low	30000.00	230000	2020-02-12 00:00:00
Bharti Airtel	Telecom	20	500000	100	Medium	75000.00	575000	2021-06-06 00:00:00
Larsen & Toubro	Infrastructure	23	600000	20	High	90000.00	690000	2023-02-14 00:00:00
Bajaj Finance	Finance	26	300000	10	High	45000.00	345000	2023-04-18 00:00:00
State Bank of India	Banking	29	700000	40	Medium	105000.00	805000	2023-07-23 00:00:00
Mahindra & Mahindra	Automobile	32	250000	25	Medium	37500.00	287500	2020-12-31 00:00:00

INSERT INTO MUTUAL_FUNDS

INSERT INTO MUTUAL_FUNDS (fund_name, fund_type, fund_manager, investment_id, invested_amount, no_of_units, risk_level, returns, current_value, date_of_investment) VALUES
('HDFC Equity Fund', 'Equity', 'Rakesh Shah', 2, 500000.00, 2000, 'High', 12.50, 560000.00, '2019-03-11'),
('SBI Bluechip Fund', 'Bluechip', 'Anjali Mehta', 5, 600000.00, 1800, 'Moderate', 8.75, 650000.00, '2022-08-15'),
('Axis Long Term Equity Fund', 'Tax Saver', 'Vikram Joshi', 9, 250000.00, 900, 'High', 10.30, 275000.00, '2020-03-14'),
('Kotak Standard Multicap Fund', 'Multicap', 'Shalini Singh', 13, 550000.00, 1400, 'Moderate', 9.20, 610000.00, '2023-04-10'),
('ICICI Prudential Value Discovery Fund', 'Value', 'Rajesh Rao', 16, 650000.00, 1600, 'Moderate', 11.10, 720000.00, '2021-07-04'),
('Mirae Asset Large Cap Fund', 'Large Cap', 'Sneha Kapoor', 19, 900000.00, 2000, 'Moderate', 8.90, 975000.00, '2022-01-09'),
('HDFC Balanced Advantage Fund', 'Balanced', 'Priya Verma', 25, 800000.00, 1500, 'Low', 7.60, 860000.00, '2022-12-20'),
('UTI Flexi Cap Fund', 'Flexi Cap', 'Alok Tandon', 28, 550000.00, 1200, 'Moderate', 9.40, 600000.00, '2021-06-16'),
('Franklin India Smaller Companies Fund', 'Small Cap', 'Meera Nair', 31, 600000.00, 1700, 'High', 12.00, 670000.00, '2022-09-11'),
('Nippon India Growth Fund', 'Mid Cap', 'Kiran Rao', 34, 300000.00, 900, 'Moderate', 8.30, 325000.00, '2022-07-01'),
('Canara Robeco Emerging Equities', 'Equity', 'Sachin Kumar', 37, 900000.00, 2000, 'High', 10.00, 990000.00, '2021-11-09'),

('DSP Tax Saver Fund', 'Tax Saver', 'Varun Desai', 40, 800000.00, 1700, 'Moderate', 7.80, 860000.00, '2022-05-14'),
('IDFC Core Equity Fund', 'Core Equity', 'Nidhi Agarwal', 46, 900000.00, 1800, 'High', 11.50, 985000.00, '2023-07-30'),
('Principal Emerging Bluechip Fund', 'Emerging Bluechip', 'Rajat Mehra', 51, 100000.00, 350, 'High', 12.75, 112750.00, '2023-01-14'),
('Sundaram Large Cap Fund', 'Large Cap', 'Nishant Jain', 55, 200000.00, 600, 'Moderate', 9.60, 219200.00, '2022-08-19'),
('Tata Mid Cap Growth Fund', 'Mid Cap', 'Kavita Rathi', 61, 150000.00, 450, 'Moderate', 10.20, 165300.00, '2021-05-21'),
('LIC MF Index Fund', 'Index Fund', 'Harish Chandra', 73, 150000.00, 500, 'Low', 6.50, 159750.00, '2022-09-09'),
('Motilal Oswal Focused 25 Fund', 'Focused', 'Amit Kapoor', 75, 300000.00, 1000, 'High', 13.20, 339600.00, '2022-11-17'),
('Aditya Birla Sun Life Equity Fund', 'Equity', 'Siddharth Shukla', 79, 150000.00, 450, 'Moderate', 8.10, 162150.00, '2023-04-10'),
('Invesco India Contra Fund', 'Contra', 'Arjun Sinha', 80, 500000.00, 1250, 'High', 9.75, 547500.00, '2021-09-22');

OUTPUT:

fund_name	fund_type	fund_manager	investment_id	invested_amount	no_of_units	risk_level	returns	current_value	date_of_investment
HDFC Equity Fund	Equity	Rakesh Shah	2	500000	2000	High	12.50	560000	2019-03-11 00:00:00
SBI Bluechip Fund	Bluechip	Anjali Mehta	5	600000	1800	Moderate	8.75	650000	2022-08-15 00:00:00
Axis Long Term Equity Fund	Tax Saver	Vikram Joshi	9	250000	900	High	10.30	275000	2020-03-14 00:00:00
Kotak Standard Multicap Fund	Multicap	Shalini Singh	13	550000	1400	Moderate	9.20	610000	2023-04-10 00:00:00
ICICI Prudential Value Discov...	Value	Rajesh Rao	16	650000	1600	Moderate	11.10	720000	2021-07-04 00:00:00
Mirae Asset Large Cap Fund	Large Cap	Sneha Kapoor	19	900000	2000	Moderate	8.90	975000	2022-01-09 00:00:00
HDFC Balanced Advantage F...	Balanced	Priya Verma	25	800000	1500	Low	7.60	860000	2022-12-20 00:00:00
UTI Flexi Cap Fund	Flexi Cap	Alok Tandon	28	550000	1200	Moderate	9.40	600000	2021-06-16 00:00:00
Franklin India Smaller Compan...	Small Cap	Meera Nair	31	600000	1700	High	12.00	670000	2022-09-11 00:00:00
Nippon India Growth Fund	Mid Cap	Kiran Rao	34	300000	900	Moderate	8.30	325000	2022-07-01 00:00:00
Canara Robeco Emerging Equ...	Equity	Sachin Kumar	37	900000	2000	High	10.00	990000	2021-11-09 00:00:00
DSP Tax Saver Fund	Tax Saver	Varun Desai	40	800000	1700	Moderate	7.80	860000	2022-05-14 00:00:00

INSERT INTO BONDS

INSERT INTO BONDS (bond_id, bond_name, investment_id, invested_amount, tenure_in_months, interest_rate, risk_level, created_at, maturity_date)
VALUES
(1, 'Sovereign Gold Bond', 7, 450000.00, 60, 7.50, 'Moderate', '2021-06-12', '2026-06-12'),
(2, 'Government Savings Bond', 10, 400000.00, 48, 6.80, 'Low', '2022-09-25', '2026-09-25'),
(3, 'Corporate Bond A', 15, 300000.00, 36, 8.00, 'High', '2019-02-08', '2022-02-08'),
(4, 'Sovereign Gold Bond', 21, 250000.00, 60, 7.50, 'Moderate', '2022-09-12', '2027-09-12'),
(5, 'Government Savings Bond', 24, 350000.00, 48, 6.80, 'Low', '2020-10-05', '2024-10-05'),
(6, 'Corporate Bond B', 36, 600000.00, 36, 8.20, 'High', '2020-02-25', '2023-02-25'),
(7, 'Corporate Bond C', 43, 400000.00, 36, 8.20, 'High', '2021-04-19', '2024-04-19'),

(8, 'Corporate Bond D', 48, 450000.00, 48, 7.90, 'Moderate', '2022-10-09', '2026-10-09'),
(9, 'Corporate Bond E', 66, 150000.00, 60, 7.20, 'Moderate', '2021-11-05', '2026-11-05'),
(10, 'Government Savings Bond', 71, 300000.00, 60, 7.00, 'Low', '2023-03-25', '2028-03-25');

OUTPUT:

bond_id	bond_name	investment_id	invested_amount	tenure_in_months	interest_rate	risk_level	created_at	maturity_date
1	Sovereign Gold Bond	7	450000.00	60	7.50	Moderate	2021-06-12 00:00:00	2026-06-12 00:00:00
2	Government Savings Bond	10	400000.00	48	6.80	Low	2022-09-25 00:00:00	2026-09-25 00:00:00
3	Corporate Bond A	15	300000.00	36	8.00	High	2019-02-08 00:00:00	2022-02-08 00:00:00
4	Sovereign Gold Bond	21	250000.00	60	7.50	Moderate	2022-09-12 00:00:00	2027-09-12 00:00:00
5	Government Savings Bond	24	350000.00	48	6.80	Low	2020-10-05 00:00:00	2024-10-05 00:00:00
6	Corporate Bond B	36	600000.00	36	8.20	High	2020-02-25 00:00:00	2023-02-25 00:00:00
7	Corporate Bond C	43	400000.00	36	8.20	High	2021-04-19 00:00:00	2024-04-19 00:00:00
8	Corporate Bond D	48	450000.00	48	7.90	Moderate	2022-10-09 00:00:00	2026-10-09 00:00:00
9	Corporate Bond E	66	150000.00	60	7.20	Moderate	2021-11-05 00:00:00	2026-11-05 00:00:00
10	Government Savings Bond	71	300000.00	60	7.00	Low	2023-03-25 00:00:00	2028-03-25 00:00:00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

INSERT INTO FIXED_DEPOSITS

INSERT INTO FIXED_DEPOSITS (fd_id, bank_name, investment_id, amount_invested, tenure_in_months, interest_rate, risk_level, created_at, maturity_date)
VALUES
(1, 'State Bank of India', 3, 750000.00, 60, 6.50, 'Low', '2020-07-18', '2025-07-18'),
(2, 'HDFC Bank', 6, 200000.00, 48, 6.25, 'Low', '2021-11-05', '2025-11-05'),
(3, 'ICICI Bank', 12, 350000.00, 36, 6.00, 'Low', '2020-12-20', '2023-12-20'),
(4, 'Bank of Baroda', 18, 400000.00, 60, 6.75, 'Low', '2019-11-19', '2024-11-19'),
(5, 'Axis Bank', 22, 700000.00, 48, 6.40, 'Low', '2021-03-03', '2025-03-03'),
(6, 'Punjab National Bank', 27, 450000.00, 36, 6.30, 'Low', '2022-05-22', '2025-05-22'),
(7, 'State Bank of India', 30, 400000.00, 60, 6.50, 'Low', '2020-01-10', '2025-01-10'),
(8, 'HDFC Bank', 33, 500000.00, 48, 6.20, 'Low', '2021-08-27', '2025-08-27'),
(9, 'ICICI Bank', 39, 600000.00, 36, 6.10, 'Low', '2019-06-13', '2022-06-13'),
(10, 'Bank of Baroda', 42, 250000.00, 60, 6.50, 'Low', '2021-12-07', '2026-12-07'),
(11, 'Axis Bank', 45, 350000.00, 48, 6.25, 'Low', '2020-03-27', '2024-03-27'),
(12, 'Punjab National Bank', 49, 800000.00, 60, 6.80, 'Low', '2021-06-03', '2026-06-03'),
(13, 'State Bank of India', 53, 150000.00, 36, 6.10, 'Low', '2021-02-10', '2024-02-10'),
(14, 'HDFC Bank', 59, 150000.00, 48, 6.30, 'Low', '2020-04-12', '2024-04-12'),
(15, 'ICICI Bank', 64, 200000.00, 36, 6.20, 'Low', '2021-06-10', '2024-06-10'),
(16, 'Bank of Baroda', 68, 300000.00, 60, 6.50, 'Low', '2020-04-18', '2025-04-18'),
(17, 'Axis Bank', 72, 150000.00, 48, 6.30, 'Low', '2020-03-11', '2024-03-11'),

(18, 'Punjab National Bank', 77, 100000.00, 36, 6.00, 'Low', '2022-05-22', '2025-05-22'),
(19, 'State Bank of India', 82, 300000.00, 60, 6.40, 'Low', '2020-10-13', '2025-10-13');

OUTPUT:

fd_id	bank_name	investment_id	amount_invested	tenure_in_months	interest_rate	risk_level	created_at	maturity_date
1	State Bank of India	3	750000.00	60	6.50	Low	2020-07-18 00:00:00	2025-07-18 00:00:00
2	HDFC Bank	6	200000.00	48	6.25	Low	2021-11-05 00:00:00	2025-11-05 00:00:00
3	ICICI Bank	12	350000.00	36	6.00	Low	2020-12-20 00:00:00	2023-12-20 00:00:00
4	Bank of Baroda	18	400000.00	60	6.75	Low	2019-11-19 00:00:00	2024-11-19 00:00:00
5	Axis Bank	22	700000.00	48	6.40	Low	2021-03-03 00:00:00	2025-03-03 00:00:00
6	Punjab National Bank	27	450000.00	36	6.30	Low	2022-05-22 00:00:00	2025-05-22 00:00:00
7	State Bank of India	30	400000.00	60	6.50	Low	2020-01-10 00:00:00	2025-01-10 00:00:00
8	HDFC Bank	33	500000.00	48	6.20	Low	2021-08-27 00:00:00	2025-08-27 00:00:00
9	ICICI Bank	39	600000.00	36	6.10	Low	2019-06-13 00:00:00	2022-06-13 00:00:00
10	Bank of Baroda	42	250000.00	60	6.50	Low	2021-12-07 00:00:00	2026-12-07 00:00:00
11	Axis Bank	45	350000.00	48	6.25	Low	2020-03-27 00:00:00	2024-03-27 00:00:00
12	Punjab National Bank	49	800000.00	60	6.80	Low	2021-06-03 00:00:00	2026-06-03 00:00:00

SR. NO	AI generated Que & Ans	Updated / New Que. & Ans	Remarks
QUE 1	Fetch Investments for an Investor		
ANS	SELECT * FROM INVESTMENTS WHERE investor_id = 10;		This query is correct.
QUE 2	Find Investments in a Specific Asset Class	Find Investments in where asset class is Stocks	

ANS	SELECT * FROM INVESTMENTS WHERE asset_class = 'Mutual Funds';	SELECT * FROM INVESTMENTS WHERE asset_class = 'Stocks';	New query is to retrieve another asset class
QUE 3	Count Total Number of Investors		
ANS	SELECT COUNT(*) AS total_investors FROM INVESTOR;		**This query is correct.**
QUE 4	List All Asset Classes	List only asset class which has low risk	
ANS	SELECT DISTINCT asset_class FROM ASSETS;	select * from assets where risk_level = 'Low';	New query will give low risk asset class info
QUE 5	Display Mutual Funds by Risk Level	Display Mutual funds with invested amount more than 600000	
ANS	SELECT * FROM MUTUAL_FUNDS WHERE risk_level = 'Low';	select * from mutual_funds where invested_amount > 600000;	New query can have info based on investments
QUE 6	Find All Investments Above a Certain Amount		
ANS	SELECT * FROM INVESTMENTS WHERE amount_invested > 100000;		
QUE 7	Fetch Top 5 Highest Investments		
ANS	SELECT * FROM INVESTMENTS ORDER BY amount_invested DESC LIMIT 5;		
QUE 8	Query to Find the Most Invested Asset Class		

ANS	SELECT asset_class, SUM(amount_invested) AS total_investment FROM INVESTMENTS GROUP BY asset_class ORDER BY total_investment DESC LIMIT 1;		